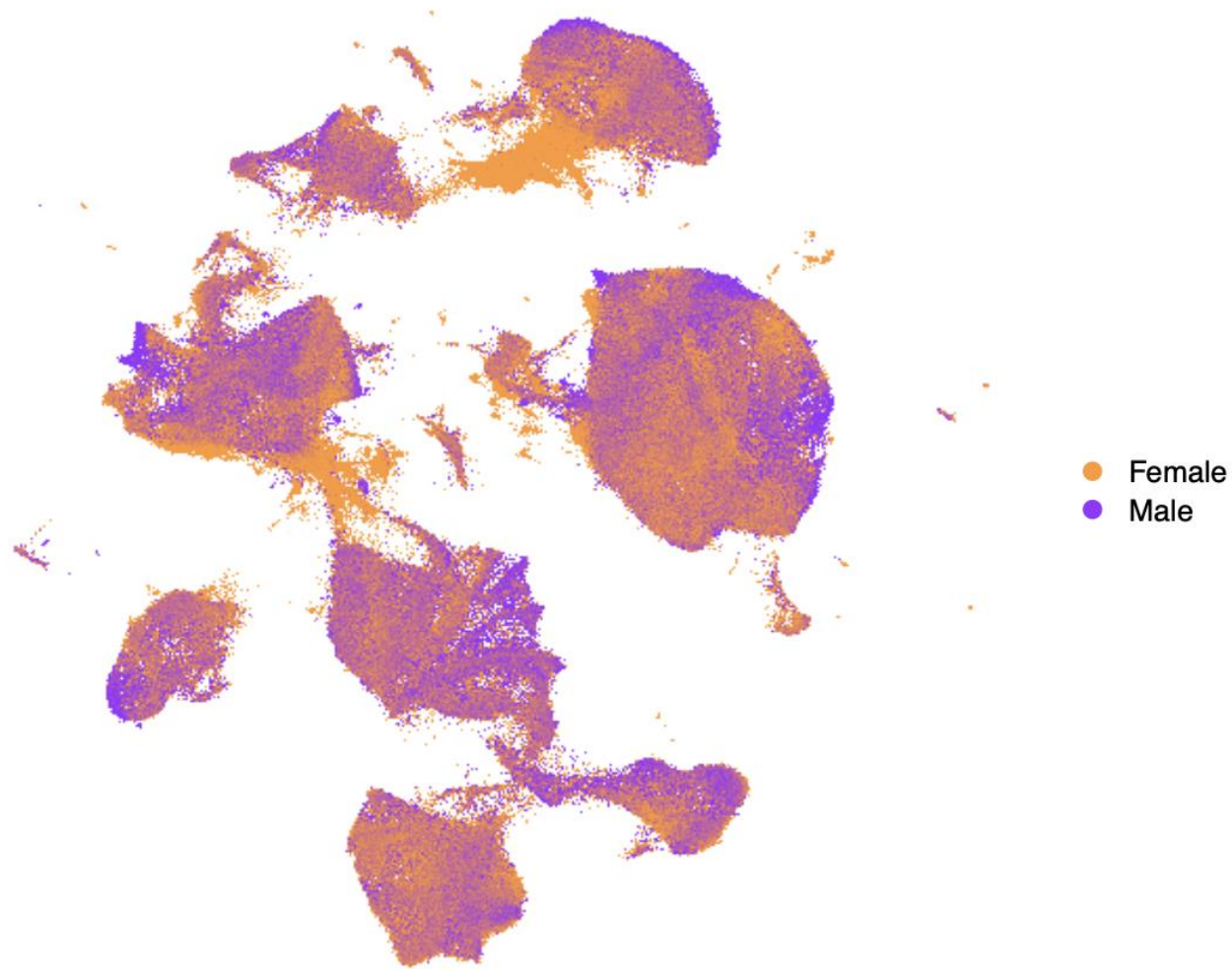


Joana P. Bernardes & Anke Fährnich
30th September 2026

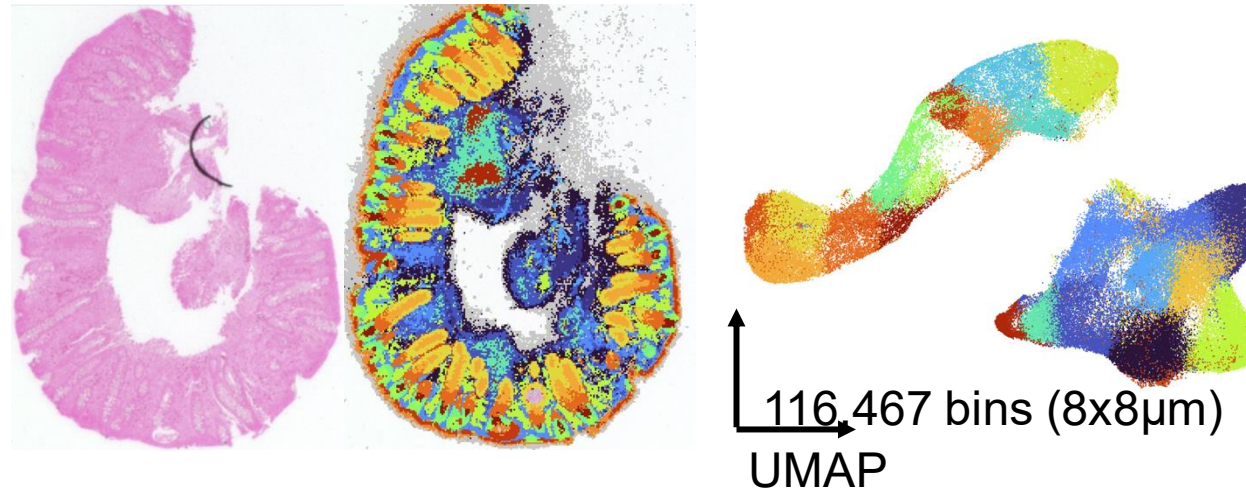
MOSAIC Summer School

Spatial transcriptomics

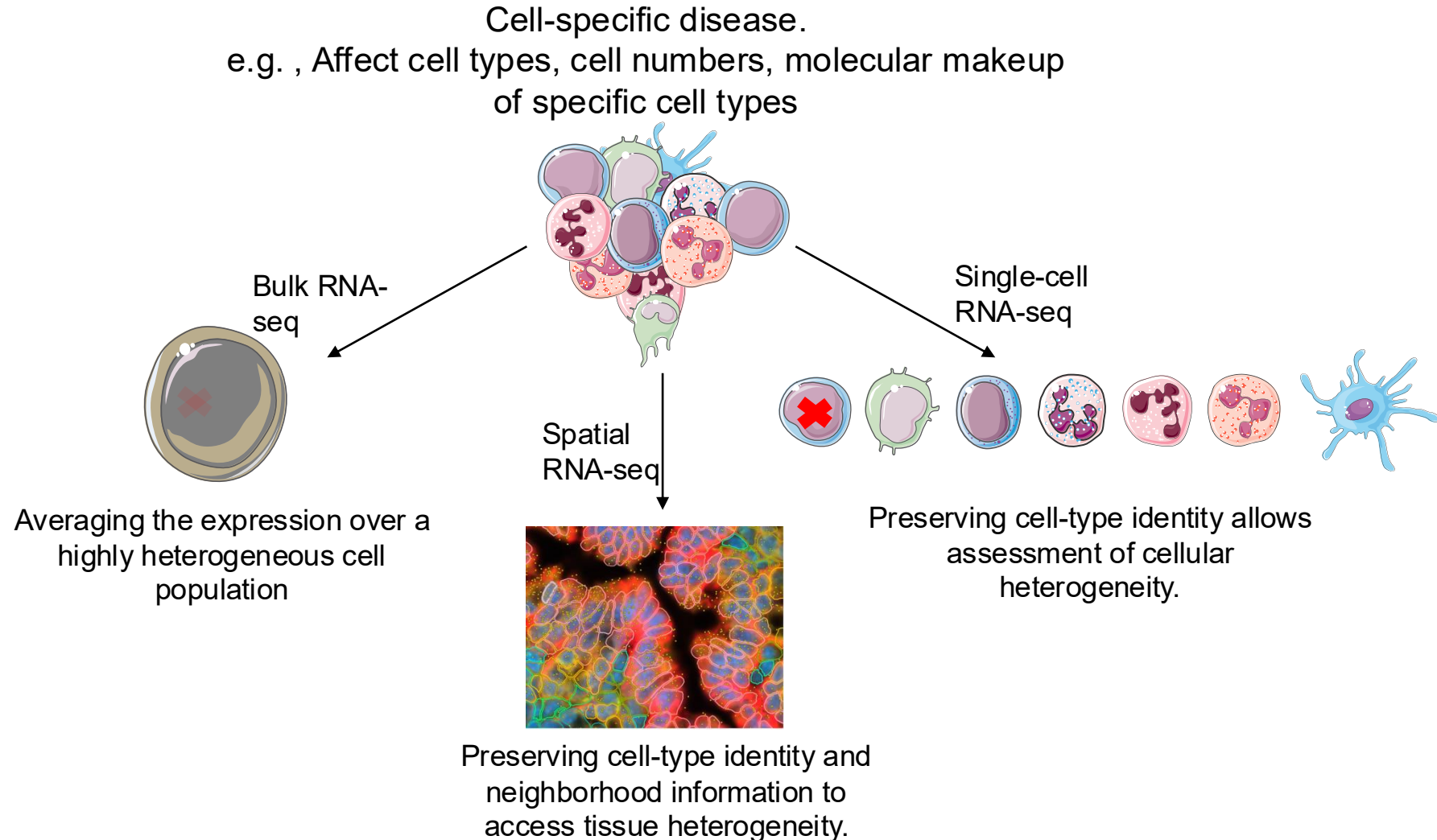


Learning objectives

- General concepts of spatial RNA-seq technology
- Technologies overview
- Spatial insights in science
- Limitations of the technology



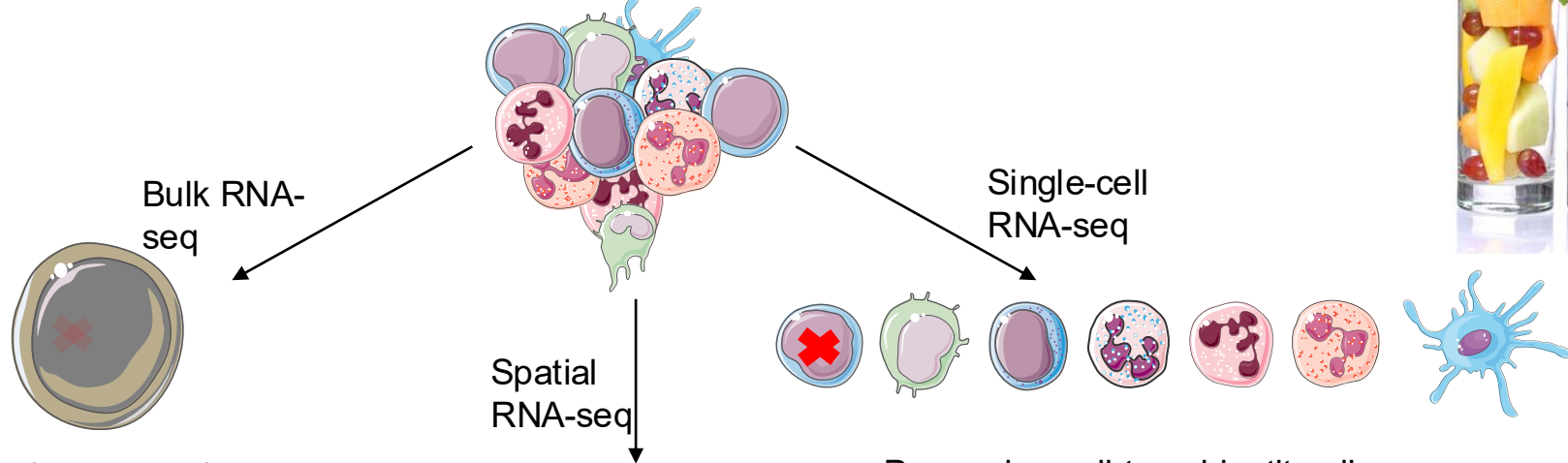
Transcriptome analysis for diseases



Something is rotten in my smoothie...



Cell-specific disease.
e.g. , Affect cell types, cell numbers, molecular makeup
of specific cell types



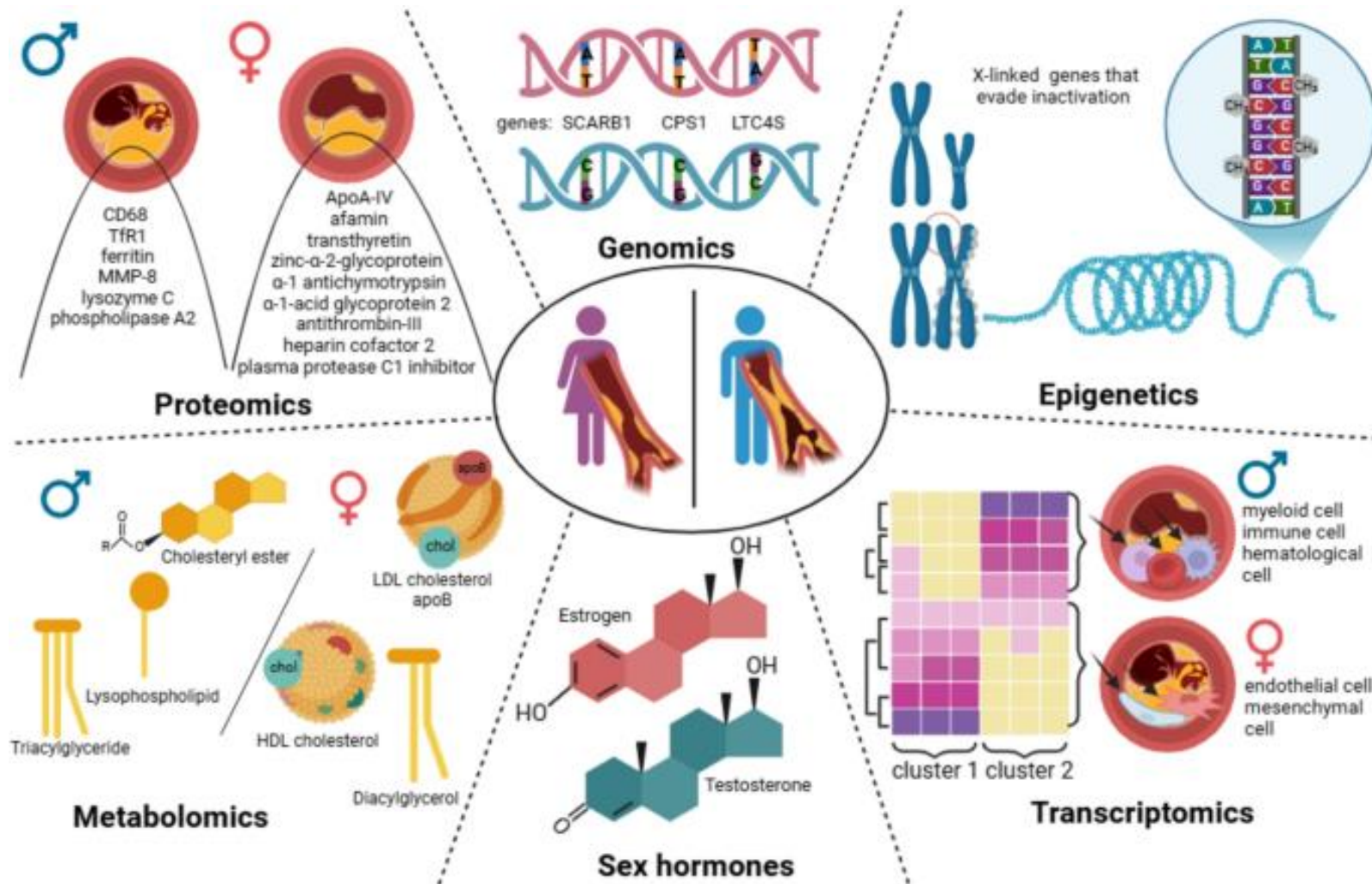
Averaging the expression over a
highly heterogeneous cell
population

Preserving cell-type identity allows
assessment of cellular
heterogeneity.

Preserving cell-type identity and
neighborhood information to
access tissue heterogeneity.

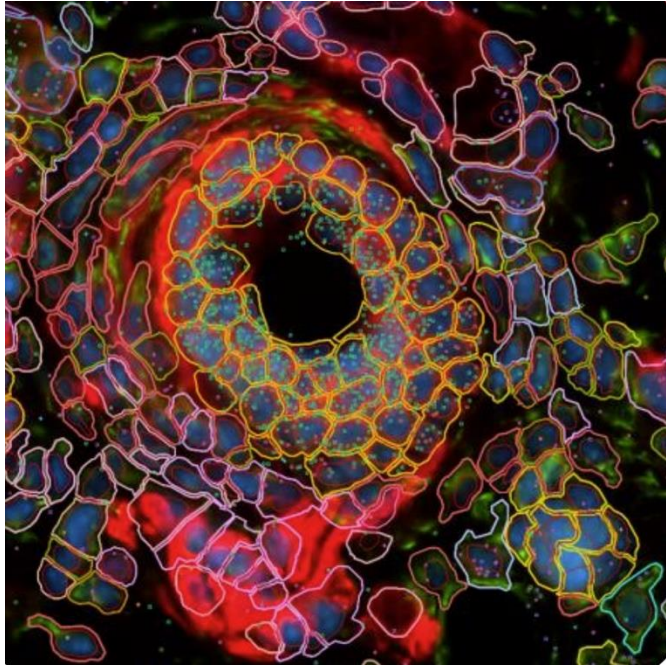


Sex sensitive perspective



Spatial Biology: Two different approaches

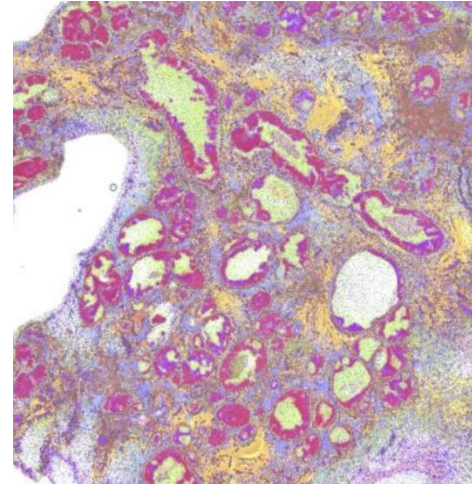
In situ hybridization



Up to 5000 RNAs & 100 proteins

**e.g. 10x Xenium
CosMx**

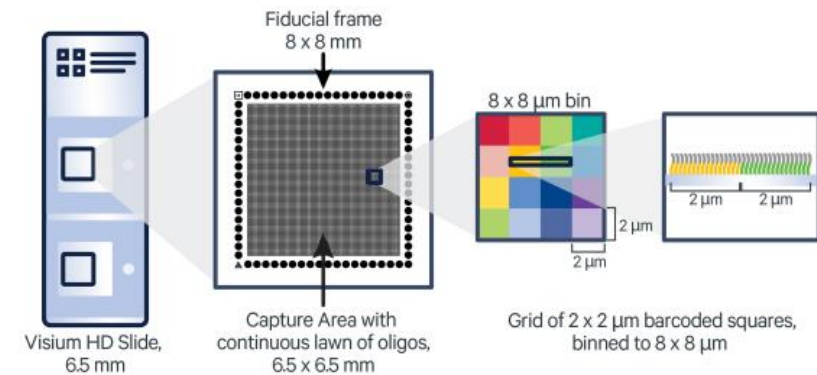
Unbiased whole transcriptome



**Probe-based solutions
18,000 genes**

or

3' Poly-A capture



**e.g. 10x Visium HD
CosMx and Illumina**

Spatial Biology: Xenium from 10x Genomics



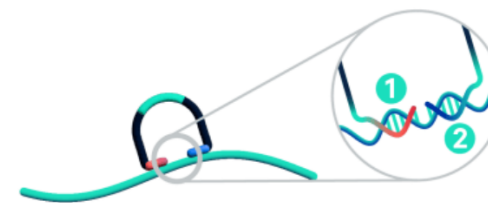
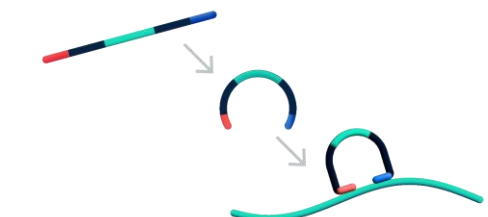
Up to 5000 RNAs & 100 proteins

Spatial Biology: Xenium from 10x Genomics



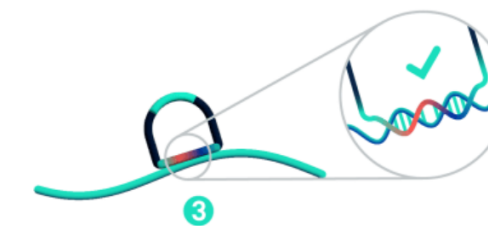
Steps 1 & 2: Dual target recognition & hybridization

Both probe arms must stably hybridize to their target. If only one arm hybridizes, the probe is unstable and will be washed off in the post hybridization wash. The padlock probes also contain a gene-specific barcode sequence that is used to generate a unique optical signature of each transcript.



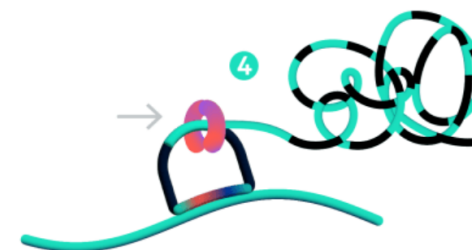
Step 3: Probe ligation

Only when both probe arms are stably hybridized to the target with a perfect match does proximal probe ligation occur. Partially bound or mis-matching probes are not ligated and cannot be amplified.



Step 4: Rolling circle amplification (RCA)

Only ligated probes are amplified, enriching for true target recognition events and suppressing off-target events. The RCA approach creates a robust and strong signal allowing single probe detection.



Xenium characteristic

- Capture area:

~10.45 × 22.45 mm (≈233 mm²)

Visium HD capture area was ~6.5 × 6.5 mm (≈42 mm²) and we added 4 biopsy cuts

Up to 20 cuts in Xenium capture area

Up to 4 Swiss rolls in the capture area

- Pannels:

Human: colon 500 and Prime 5K panel

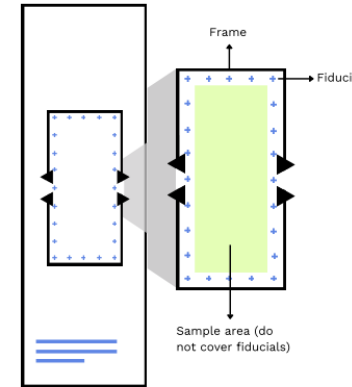
Mouse: Prime 5K panel and brain and small tissue

First trial human: 500 colon panel + 51-100 genes

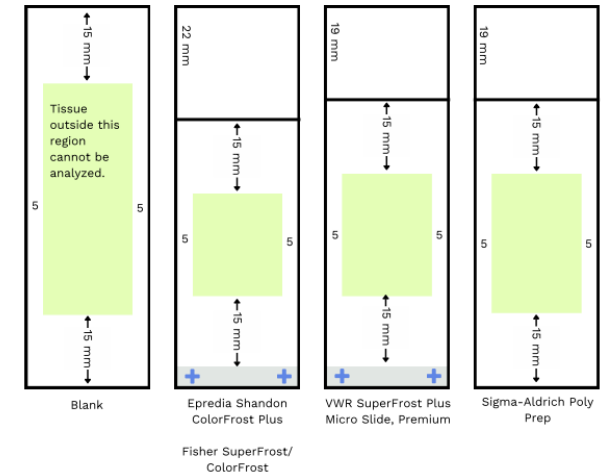
First trial mouse: 5k tissue and pathways panel

Protein panel now available (up to 27 proteins)

Xenium



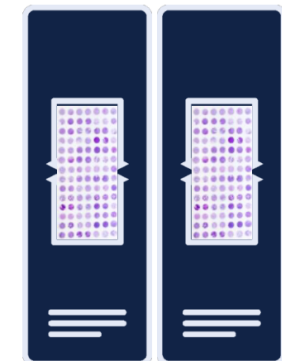
Visium HD



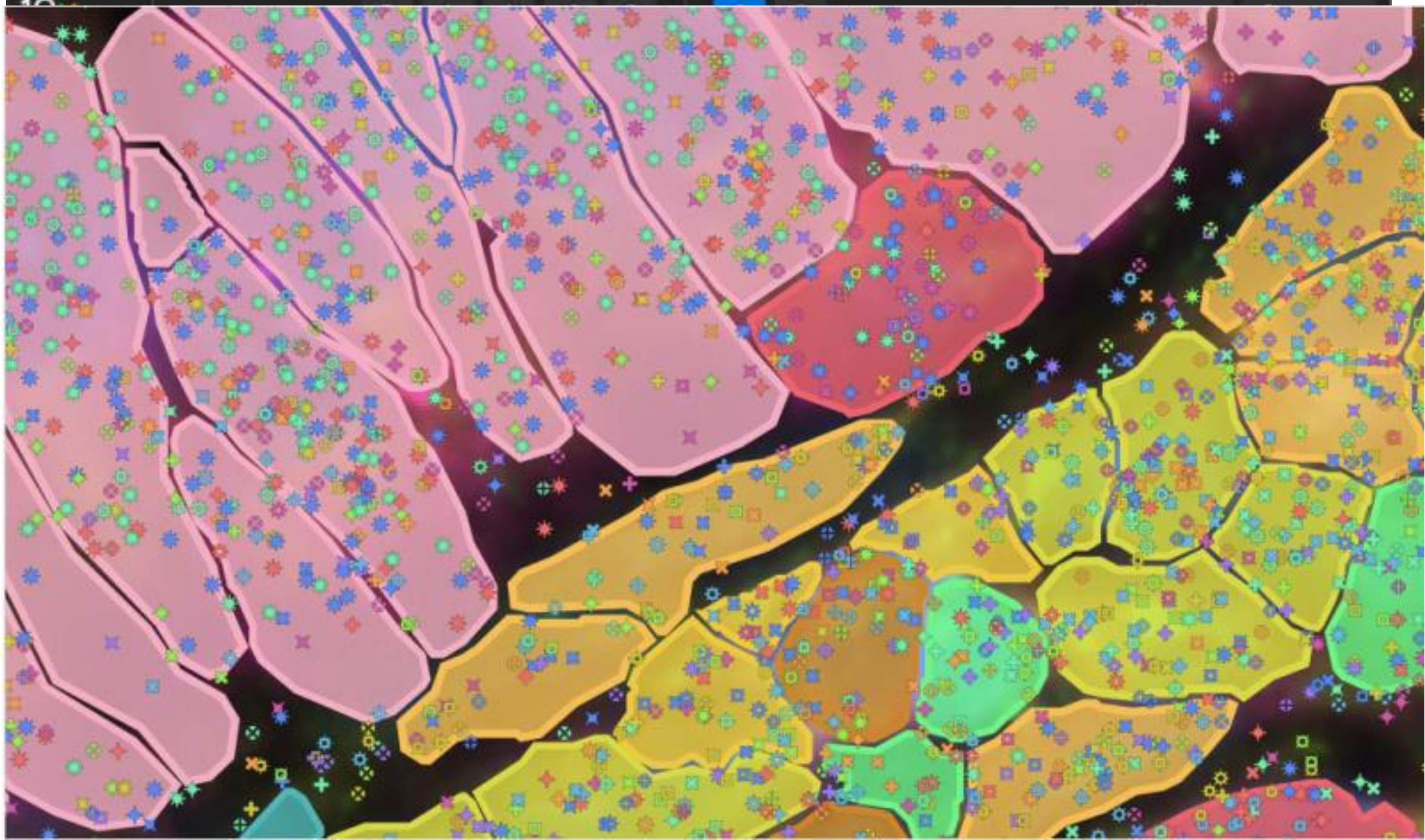
Tissue must have 5 mm margin on each side.
Tissue must have 15 mm margin top and bottom.



Six 6 x 10mm mouse brain sections



196 x 1mm biopsies in TMAs



Spatial Biology: CosMx from Bruker

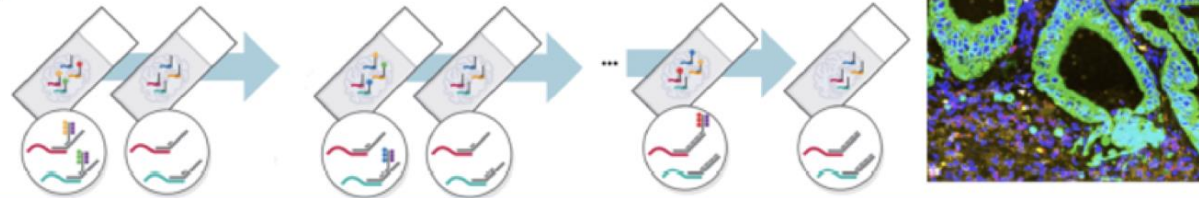
Slide Preparation

Manual Method or automated with
Leica BOND RX/RX^m



Process samples on CosMx SMI

Automated Cyclic *in situ* Hybridization
Chemistry



Interactive Data Analysis

Cloud-based scalable computing and
storage with interactive data viewer



Run time: ? (BCN colleagues); four slides per run

CosMx characteristics

- Capture area:

~300 mm²

Select ROIs (regions of interest)

Up to 20 cuts in capture area

Up to 5 Swiss rolls in the capture area

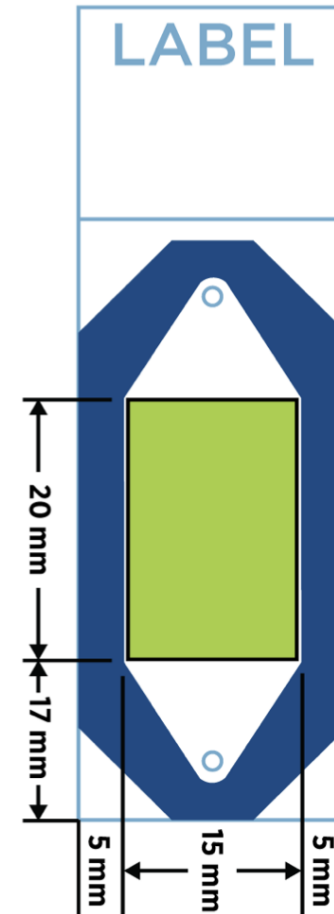
- Pannels:

Human whole transcriptome (18k)

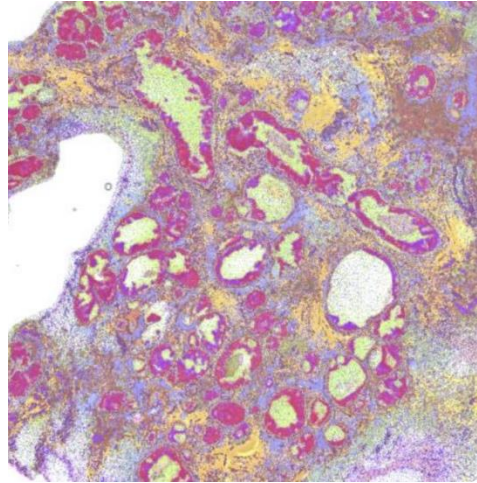
Human 6k discovery panel

Human and mouse 1k cell characterization panel

Protein panel now available (up to 72 proteins)



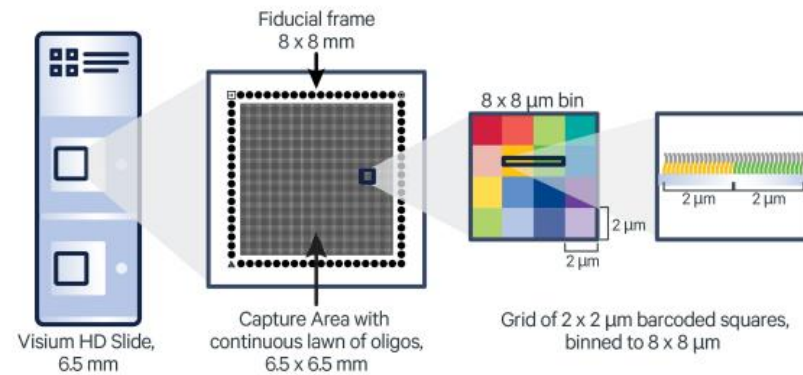
Unbiased whole transcriptome



Probe-based solutions
18,000 genes

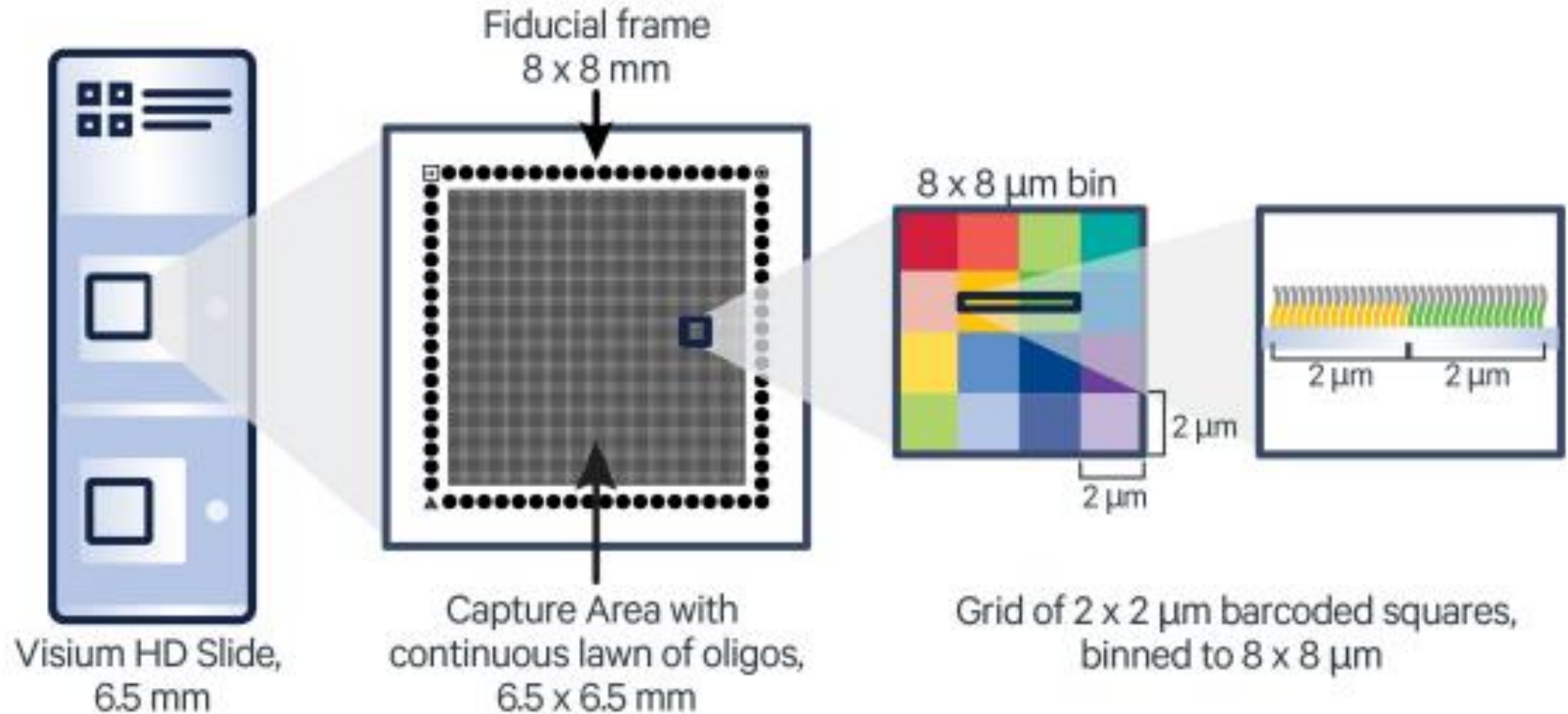
or

3' Poly-A capture



e.g. 10x Visium HD
CosMx and Illumina spatial

10x Visium HD: Increasing portfolio of application



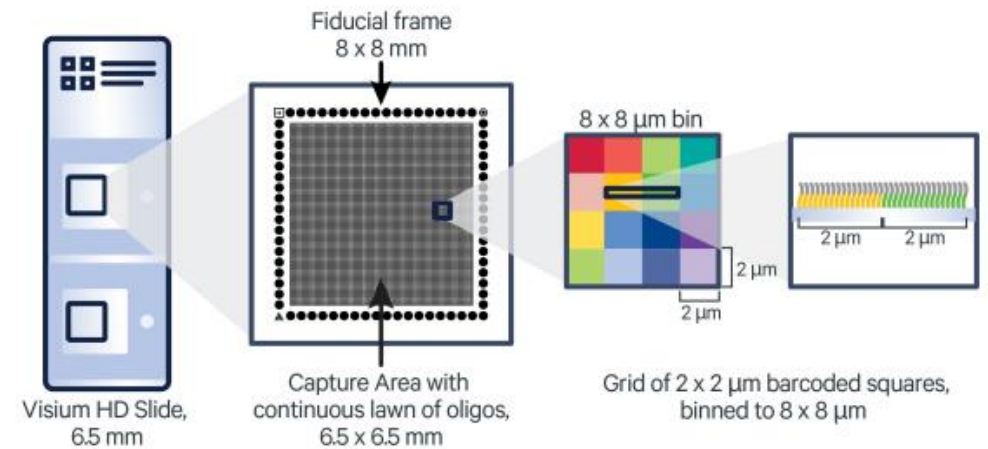
10x Visium HD: Increasing portfolio of application

HD Spatial Gene Expression

- Probe-based capture (18.000 genes)
- Human / Murine
- FFPE, Fresh Frozen, Fixed Frozen

HD 3' Spatial Gene Expression

- Whole Transcriptome
- No species restrictions
- Fresh Frozen



Types of tissue: FFPE



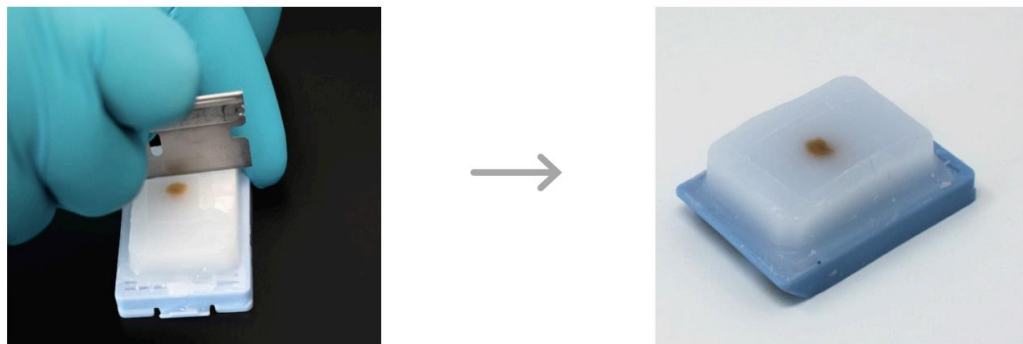
Any archived FFPE blocks
(Formalin-Fixed Paraffin-Embedded)



Tissue microarray

Mounting FFPE

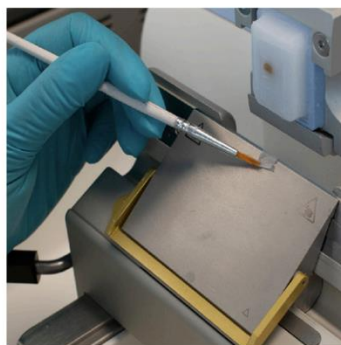
Trimming large tissue block containing tissues that fit the Capture Areas



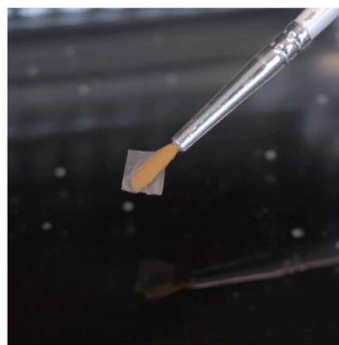
FFPE Tissue Block in Ice Bath



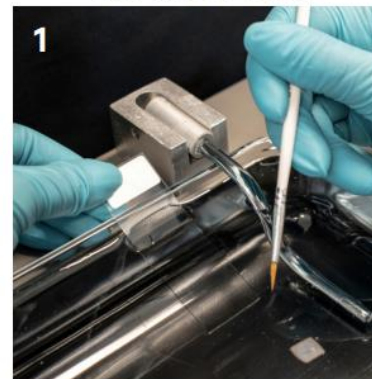
Lift the section with the paint brush



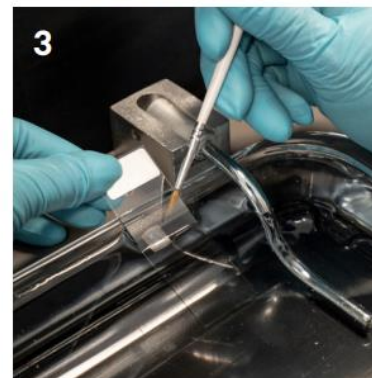
Place section in the water bath



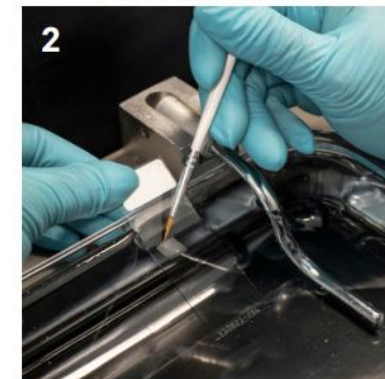
Section floating in the water bath



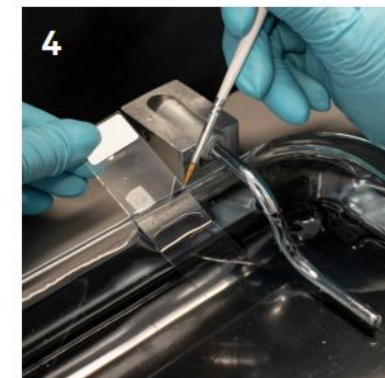
Align section edge with desired location



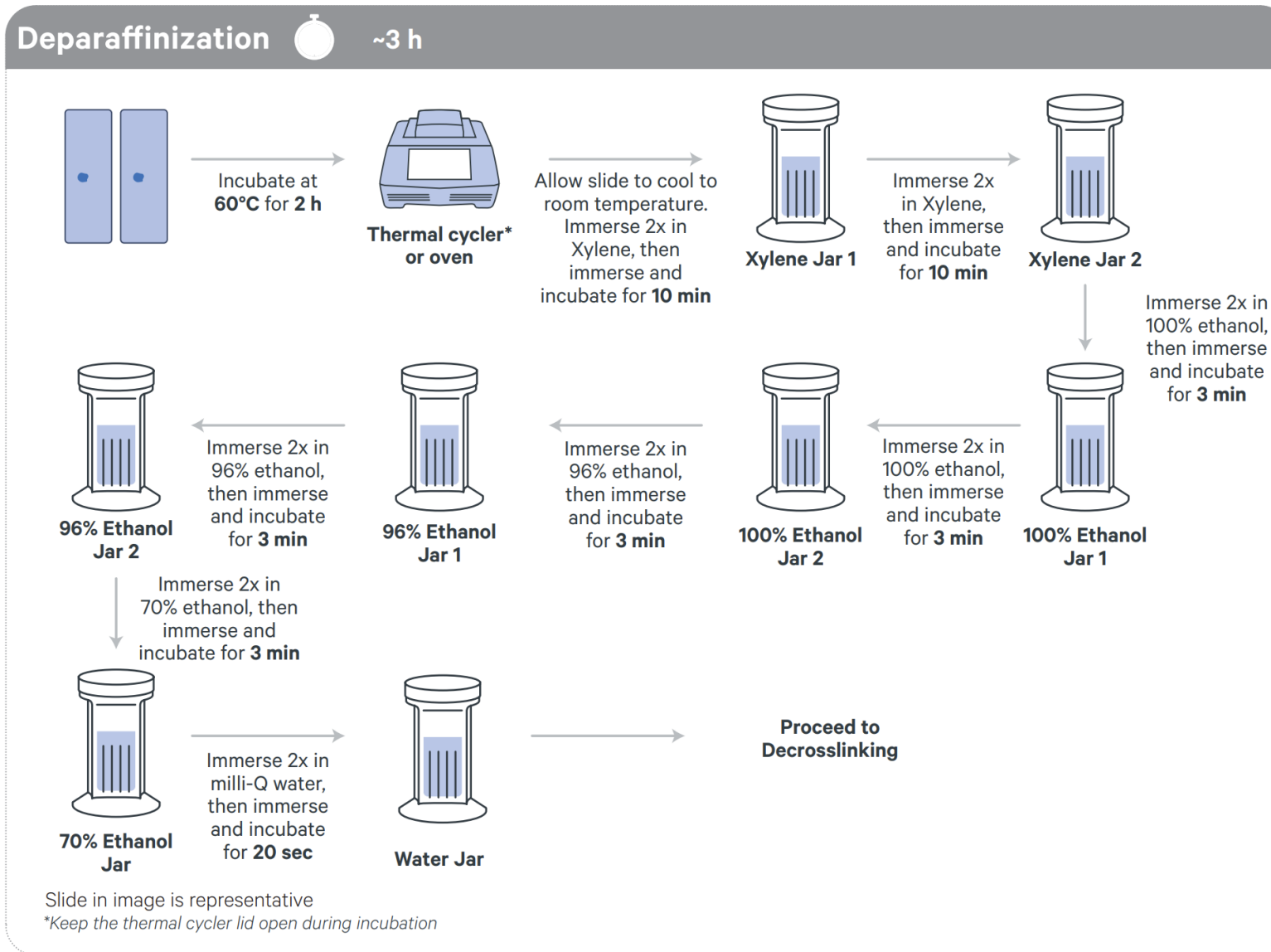
Blank slide inserted below surface of water



Pull the blank slide out of the water

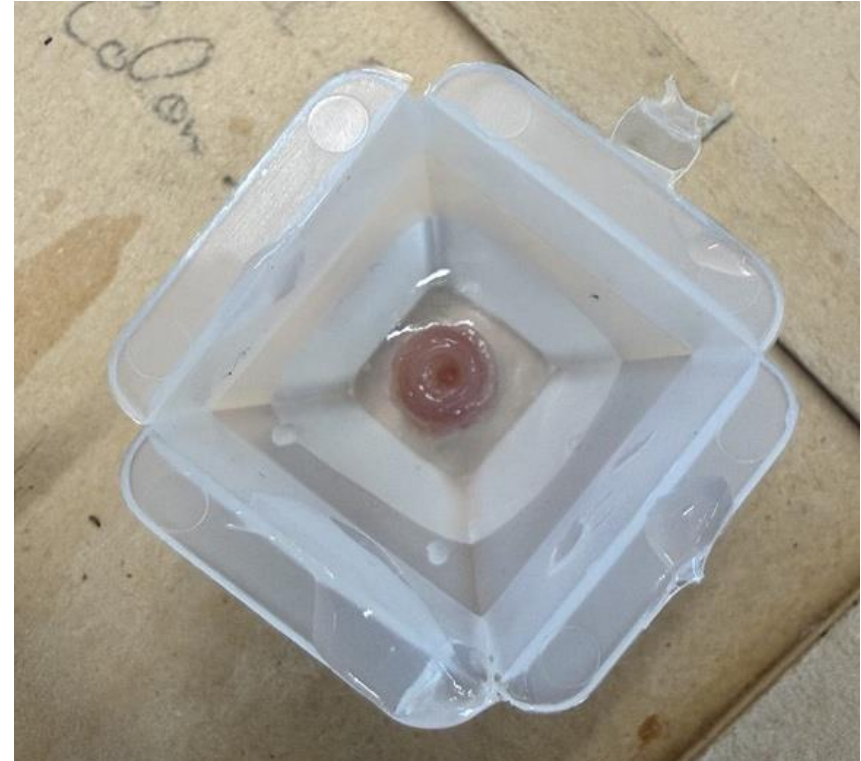


Essential steps after



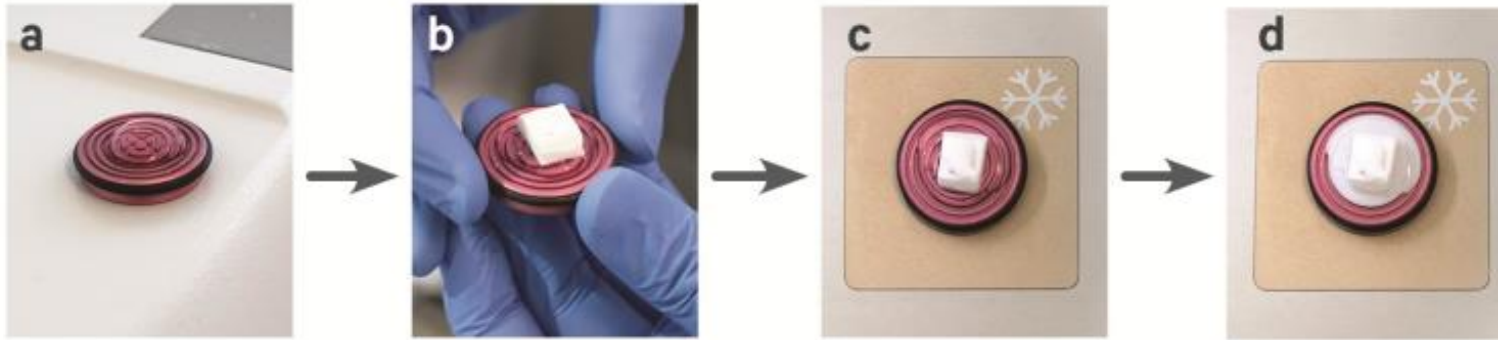
Types of tissue: Fresh frozen

Tissue OCT-embedded
Can use different Matrix types:
e.g. CMC embedding
for metabolomic acquisition



Mounting Fresh Frozen

OCT Embedded Tissue Block Mounting



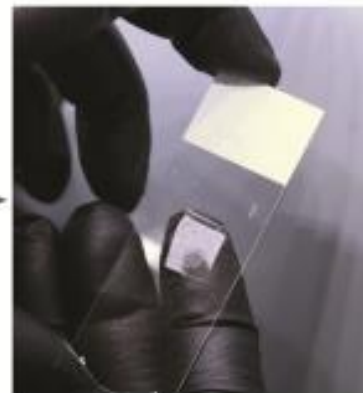
Flatten the Section



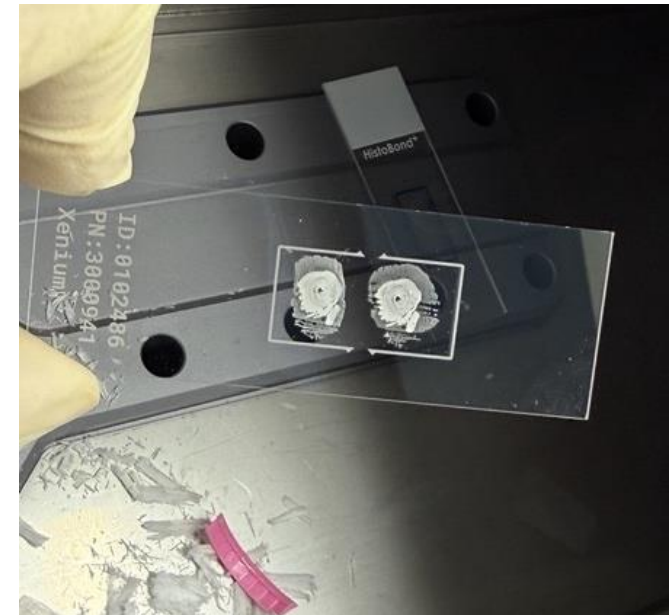
Transfer the Section



Adhere the Section

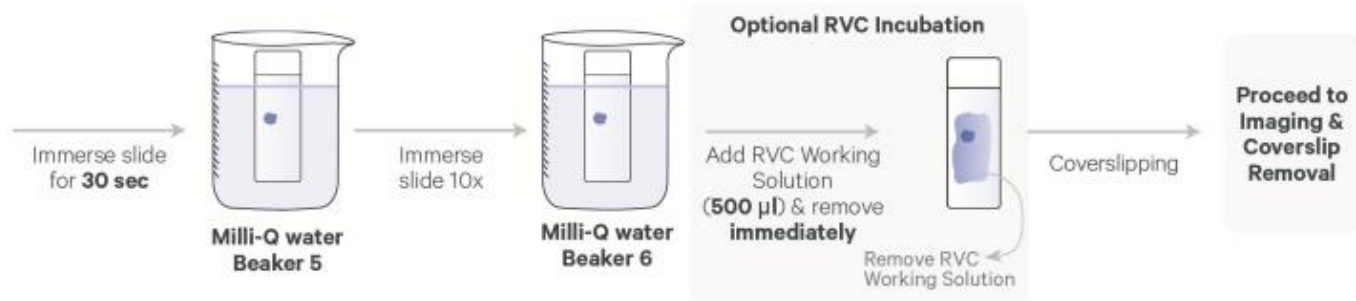
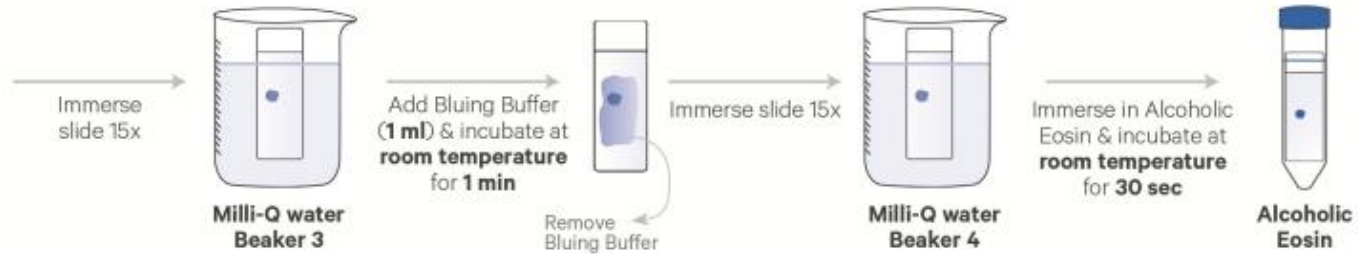
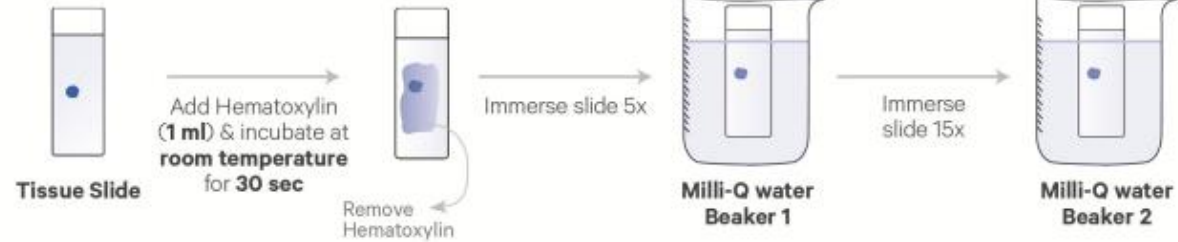


Immediately place the slide on the cryobar to allow section to freeze



Steps after Fresh Frozen

H&E Staining & Coverslipping ⌚ ~15 min



Wipe excess liquid from the back of the slide after each immersion series

Pilot trial: FFPE IBD biopsies for Visium



10x Visium CytAssist

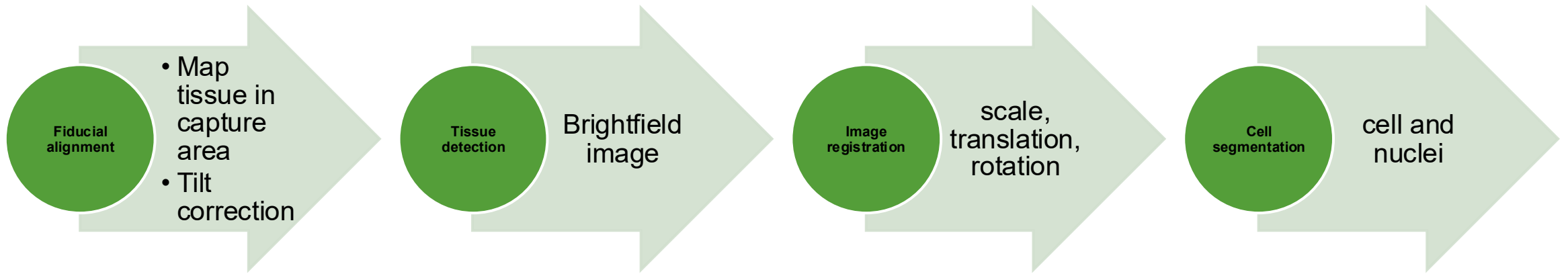
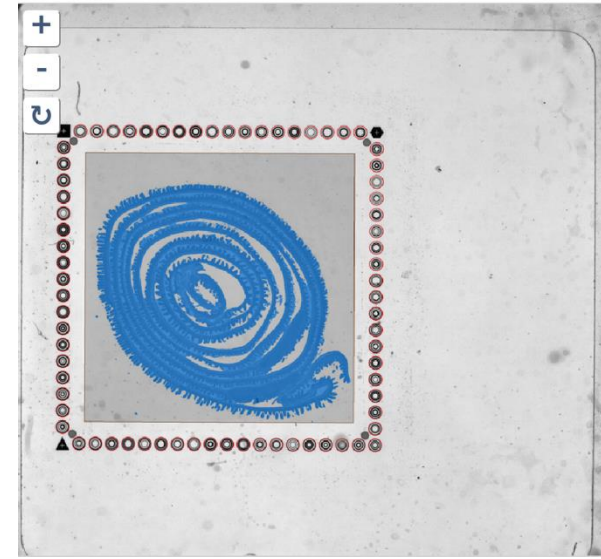


6.5mm

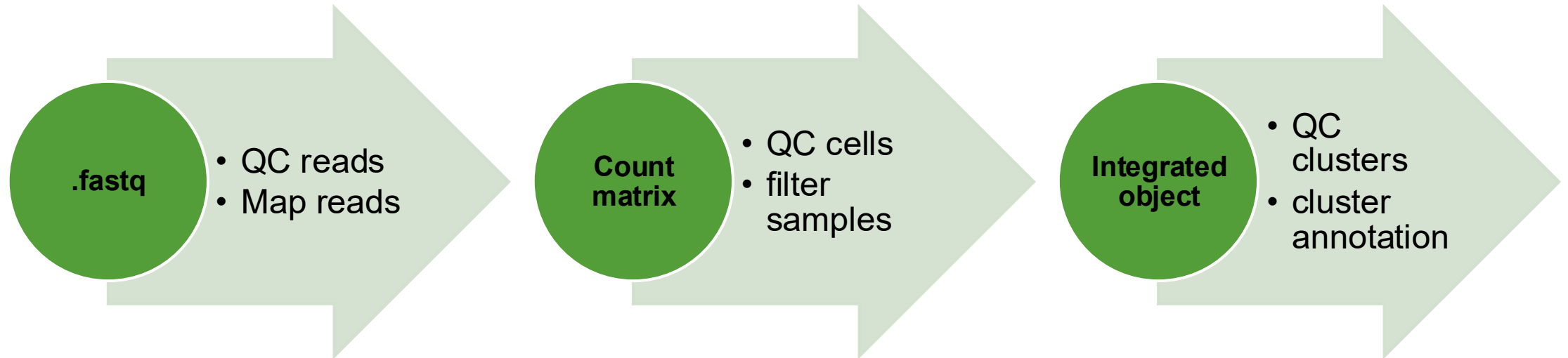


Sequencer

Get image of slide

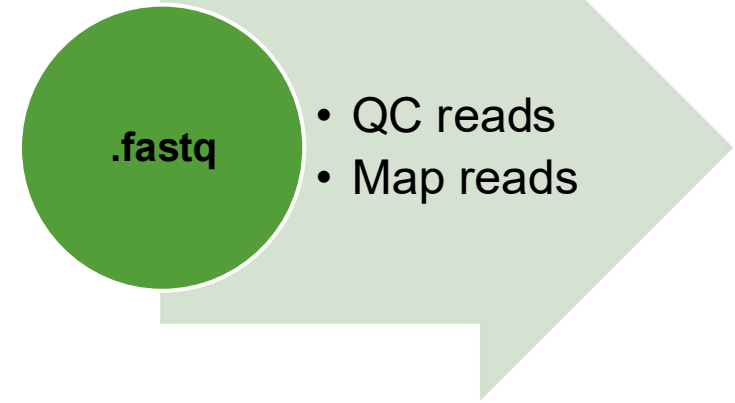


Associate locations with features (transcripts, metabolites...)



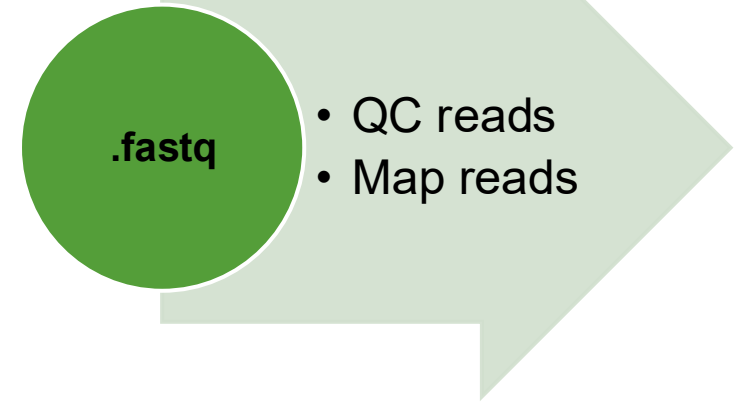
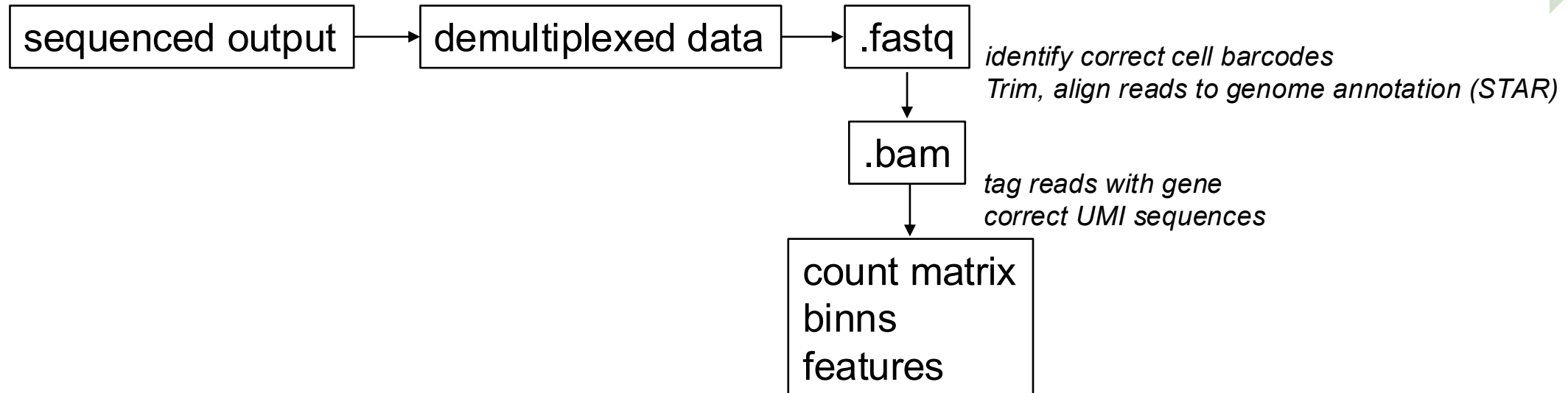
From reads to count matrix

- space ranger

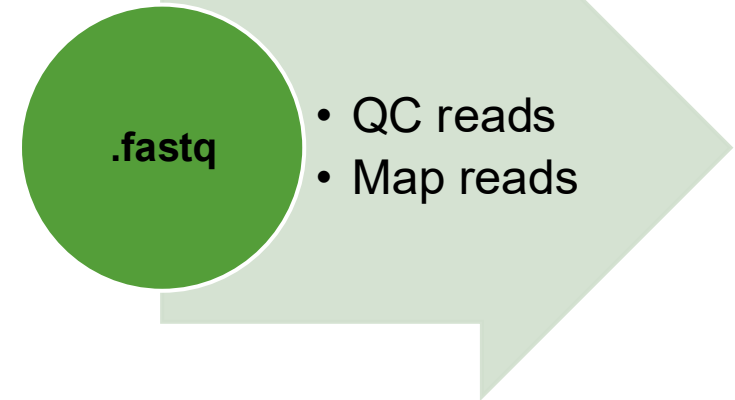


From reads to count matrix

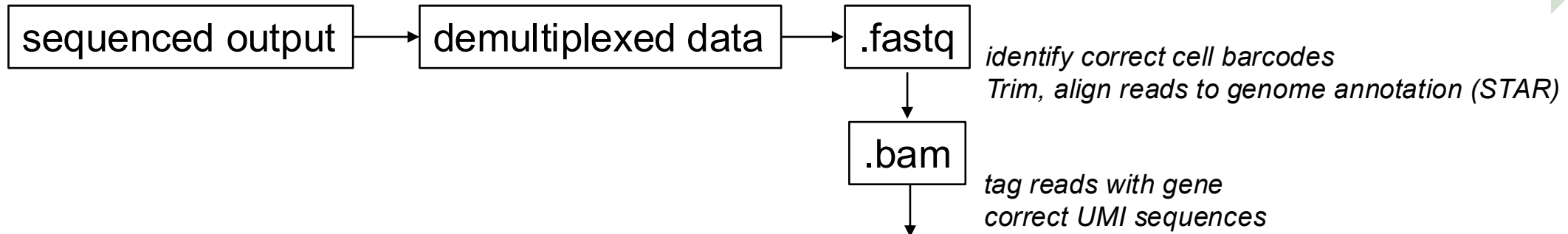
- space ranger



From reads to count matrix



- space ranger



	bin 1	bin 2	bin 3	...	bin 2M
Gene 1	3	3	1		2
Gene 2	0	2	0		1
Gene 3	1	0	3		5
...					
Gene K	14	7	1		0
...					
Gene 25K	0	13	1		0

count matrix
binns
features

- Lots of zeros due to dropout or no expression
- Variable total counts across cells – cell specific bias
- Variable counts per gene – part biological, part technical

From matrix to objects

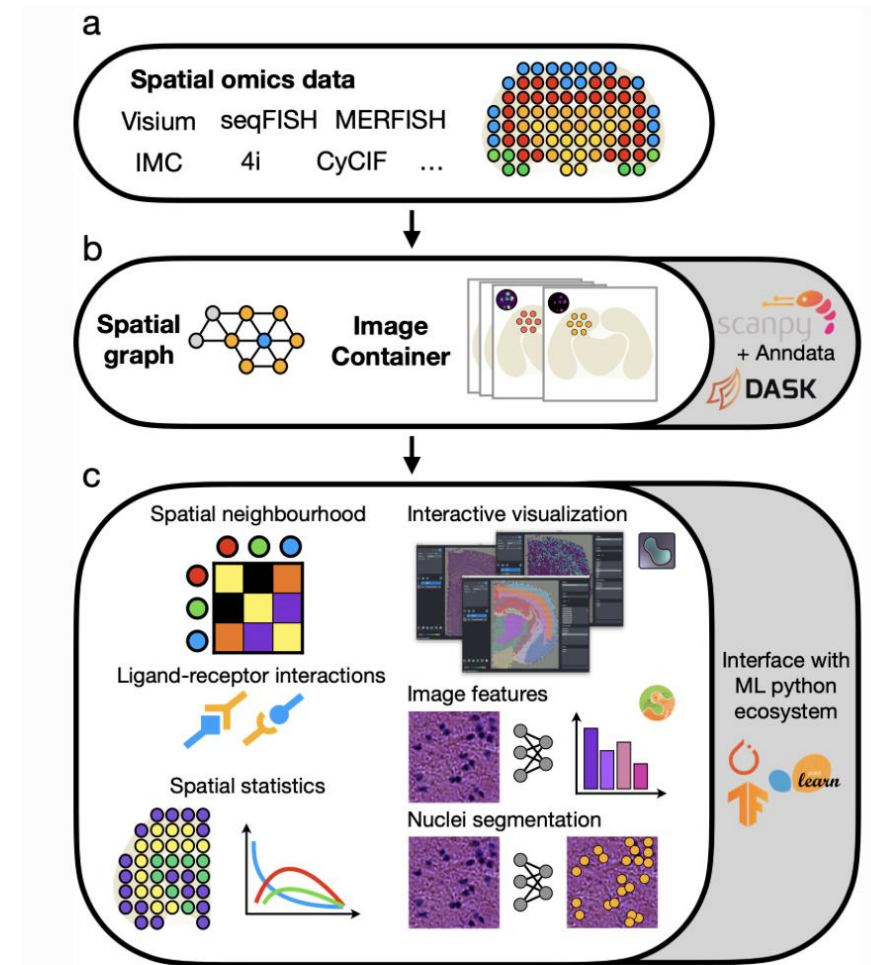
- Create object

R packages

- **Seurat V5** > **S4 seurat object**
- Banksy V1 > SingleCellExperiment

Python packages

- Squidpy V1.7.0 > Anndata
- SpaGCN 3.0.0 > h5ad



Inflammatory Bowel Disease

Crohn's disease / Ulcerative colitis

Please do not look if Images
of diseased tissues affect you

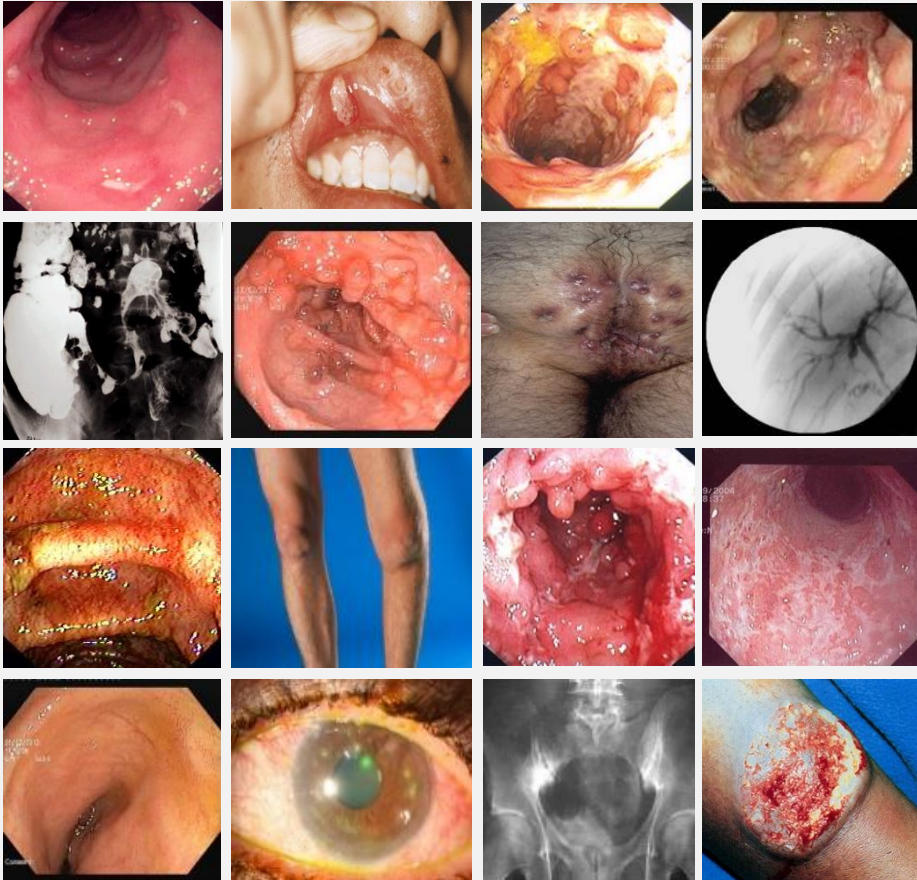
- **Common disease, compounding prevalence (ca. 1%)¹**
- **Pathogenesis:** Complex interplay of
 - Genetics²
 - Environmental triggers
 - Immune responses
- **Diagnostics and monitoring is not trivial**
- **Therapeutic responses vary (max 50% per drug)**
 - No biomarkers available

¹ Ng *et al*, 2017 Lancet

² Franke *et al*, 2010 Nat Genet

Inflammatory Bowel Disease

Crohn's disease / Ulcerative colitis

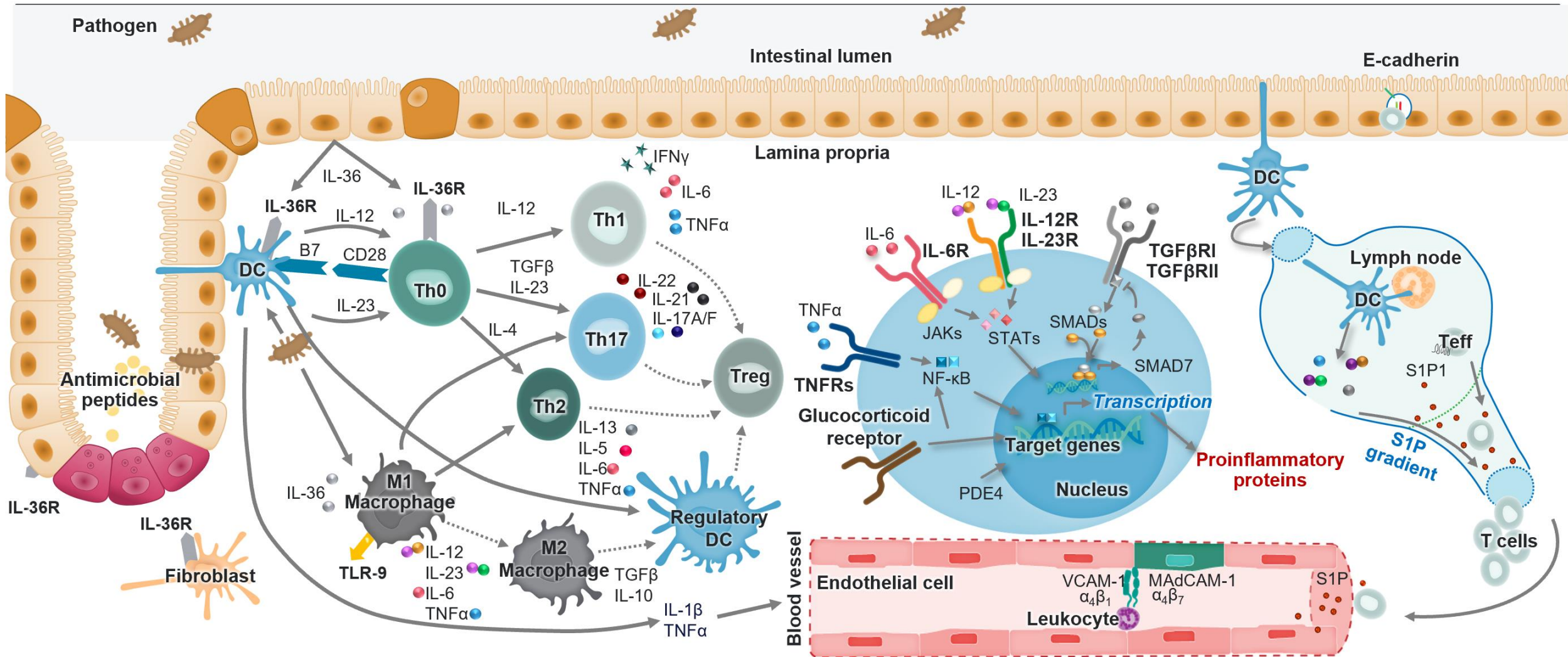


- **Common disease, compounding prevalence (ca. 1%)¹**
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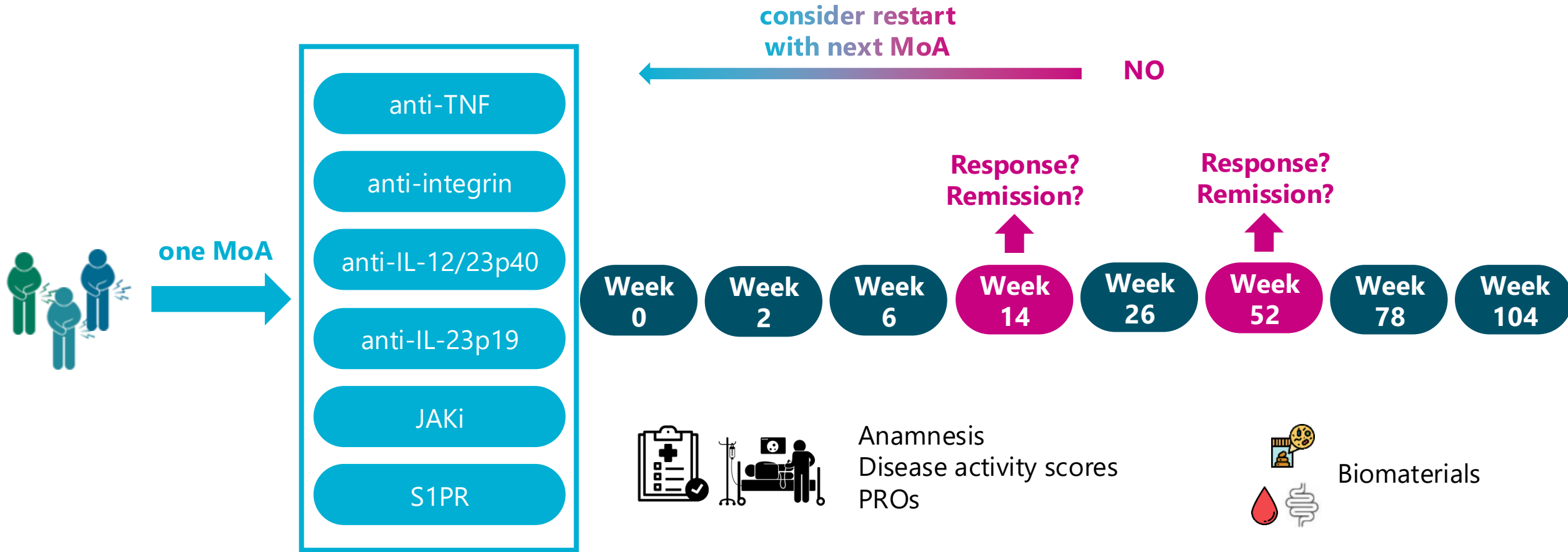
The complexity of immunopathologic pathways in IBD¹⁻⁴



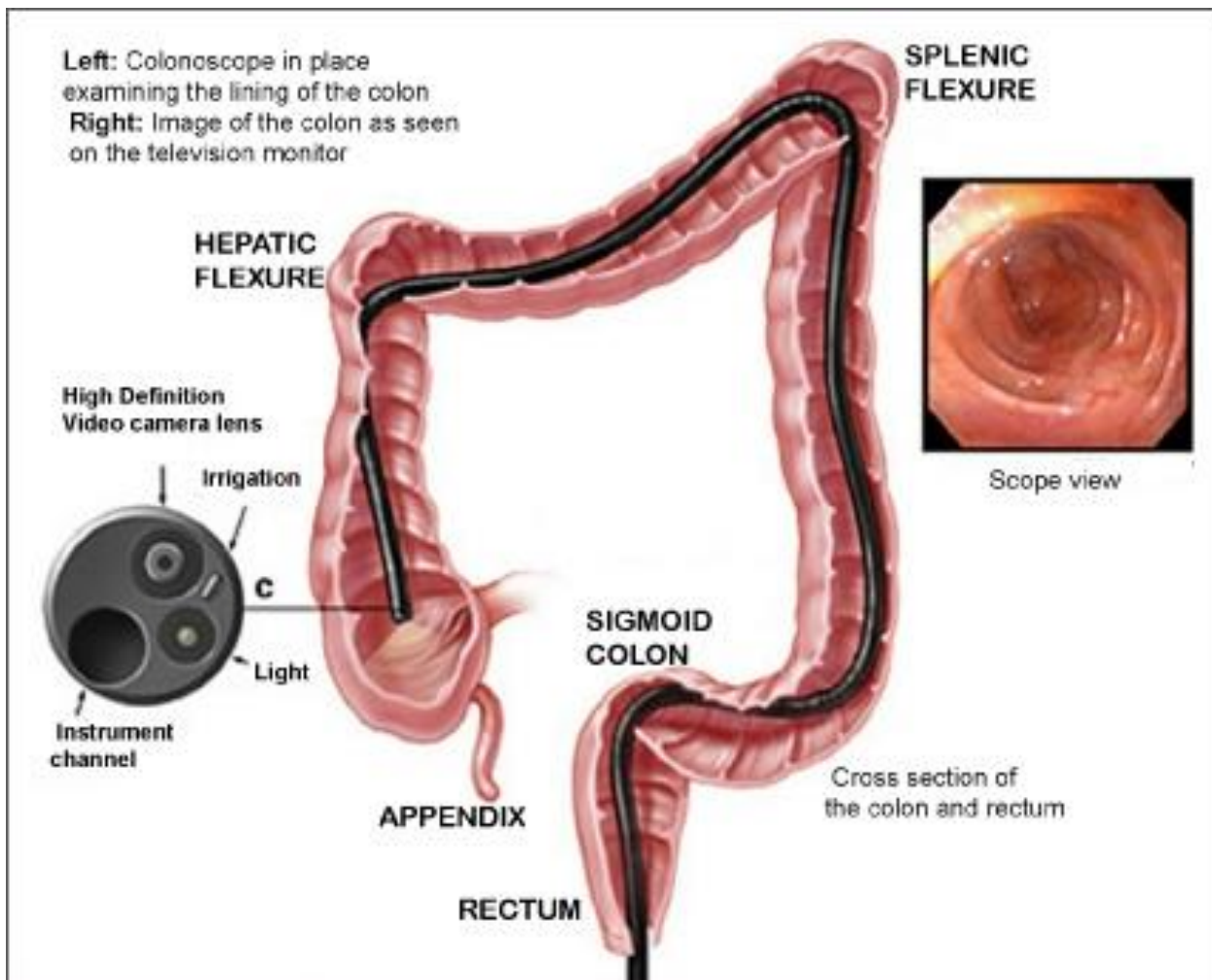
CD, cluster of differentiation; DC, dendritic cell; IFN, interferon; IL, interleukin; JAK, Janus kinase; MAdCAM-1, mucosal vascular addressin cell adhesion molecule 1; NF-κB, nuclear factor-κB; PDE4, phosphodiesterase type 4; R, receptor; S1P, sphingosine-1-phosphate; SMAD7, small mothers against decapentaplegic homolog 7; STAT, signal transducer and activator of transcription; Teff, effector T cell; TGF, transforming growth factor; Th, T helper cell; TLR, toll-like receptor; TNF, tumor necrosis factor; Treg, regulatory T cell; VCAM-1, vascular cell adhesion protein.

1. Adapted from Coskún M, et al. *Trends Pharm Sci.* 2017;38:127–42; 2. Nielsen OH, et al. *Expert Opin Investig Drugs.* 2016;25:709–18; 3. Dubois-Camacho K, et al. *World J Gastroenterol.* 2017;23:6628–38; 4. Neurath MF. *Nat Rev Immunol.* 2014;14:329–42.

Study design

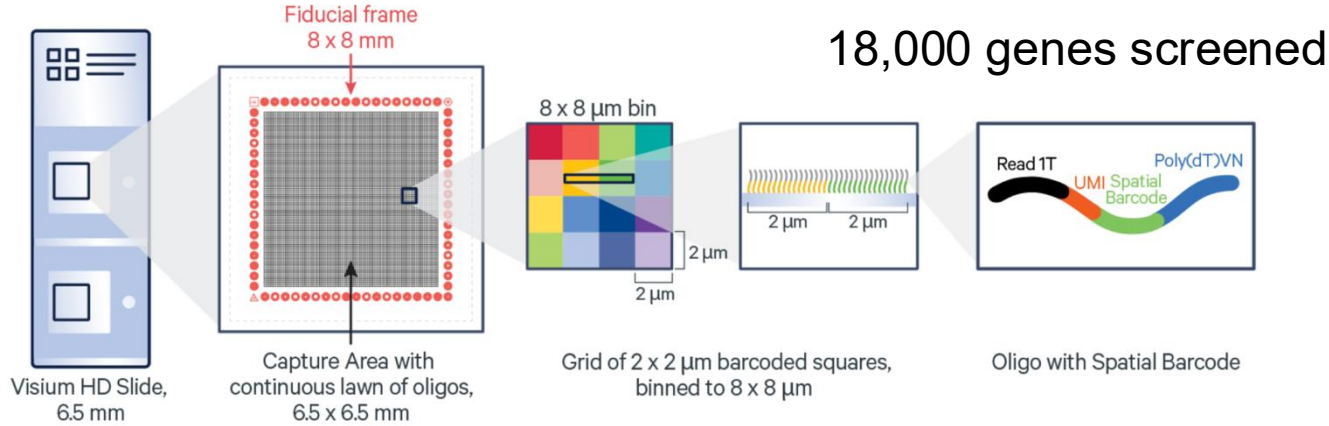


„Multicenter-like“ longitudinal observation and biomaterial trial
First patient in: October 2020

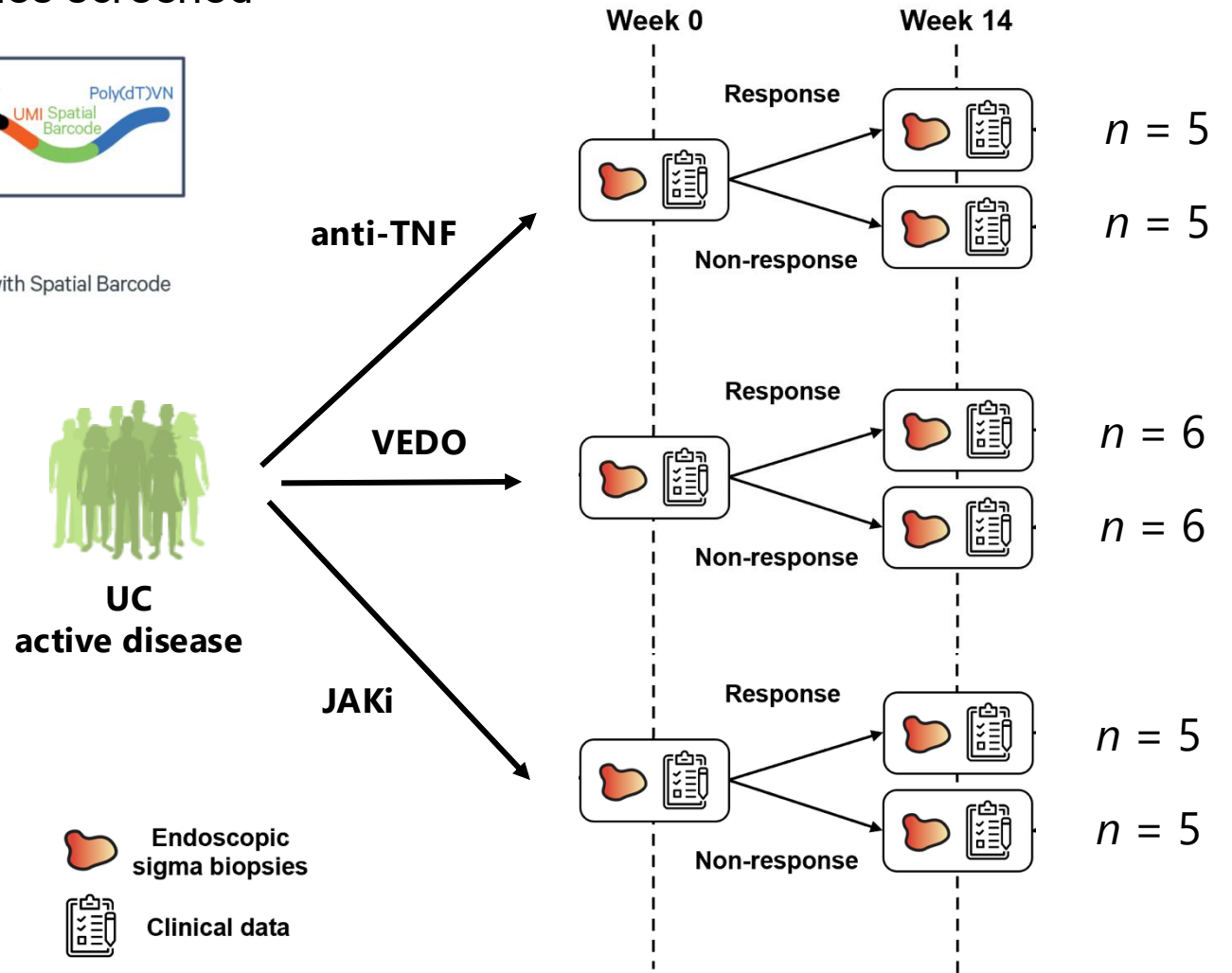


Biopsy size: 2-3 mm diameter

• Spatial transcriptomics: Aims and Experimental design



- ❖ DET at cellular level
- ❖ Immune cells localization, infiltration → Nearest neighbour analysis
- ❖ Categorised at W14 (responders vs non-responders)



Insights

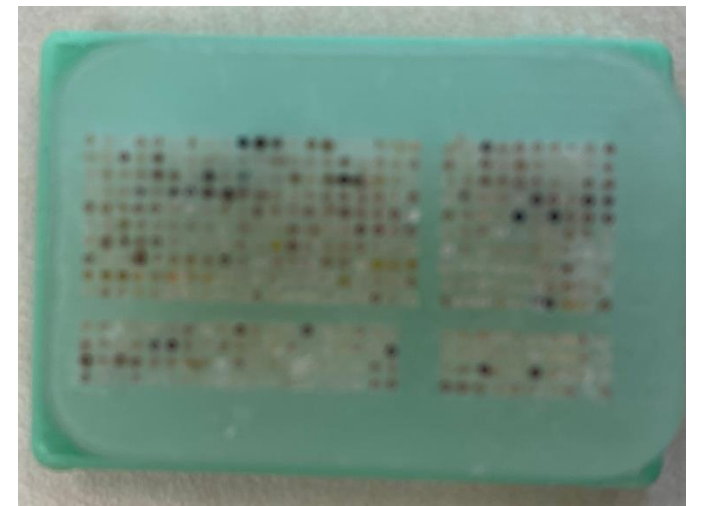
Disease- and sex specific mechanisms of disease:

What leads to CID?

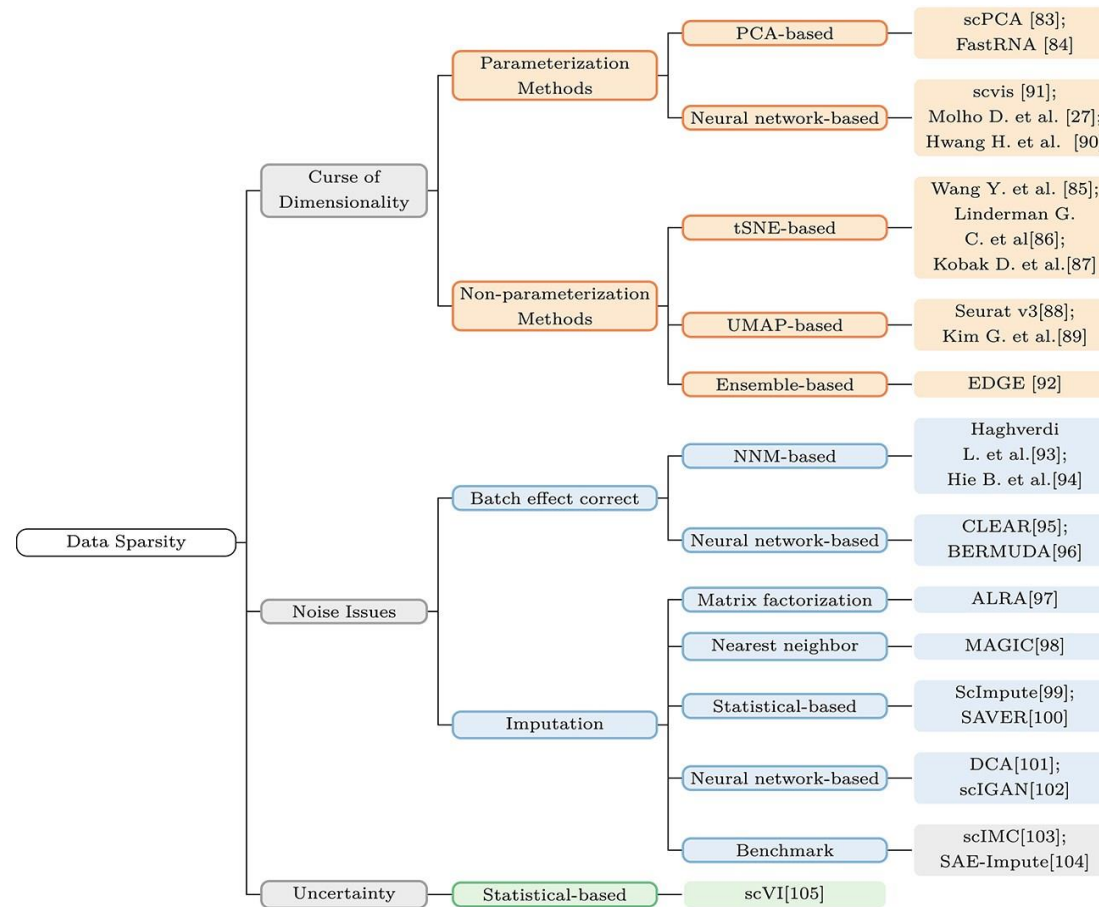
Can we predict a flare?

Can we determine whether a patient will respond to treatment before administering it?

Advantage of years of collection (FFPE)



Limitations of the technology



Limitations of the technology



Price



Tissue
collection



Time



Computational
limitations

Check hands on tutorial

- Please go to https://github.com/Systems-Immunology-IKMB/spatialRNAseq_workshop/





Any
questions?

Where can I get public data?

Best practices:

- [Single-cell best practices](<https://www.sc-best-practices.org/preamble.html>)
- [Orchestrating Single-Cell Analysis with Bioconductor](<https://bioconductor.org/books/release/OSCA/>)
- [Orchestrating Spatial Transcriptomics Analysis with Bioconductor](<https://lmweber.org/OSTA/>)
- Seurat:
 - [Essential commands](https://satijalab.org/seurat/articles/essential_commands)
 - [Basic Single-cell Analysis Vignette](https://satijalab.org/seurat/articles/pbmc3k_tutorial)
 - [Seurat Visium Analysis Vignette](https://satijalab.org/seurat/articles/spatial_vignette)
 - [Seurat Visium HD Analysis Vignette](https://satijalab.org/seurat/articles/visiumhd_analysis_vignette)
- Tools:
 - [Spaceranger](<https://www.10xgenomics.com/support/software/space-ranger/latest>)
 - [Loupe Browser](<https://www.10xgenomics.com/support/software/loupe-browser/latest>)
 - [Spotsweeper](<https://github.com/MicTott/SpotSweeper>)
 - [banksy](<https://github.com/prabhakarlabs/Banksy>)....

